PCBC H Parametric & Sequences Test (Practice C) **Name**_____ _____/27

SHOW ALL WORK FOR FULL CREDIT. ***Calculator*** **You have the entire period.**

Honor Code (copy below & sign): On my honor, before God, I have neither given, nor received any aid on this test.

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_____ _____

Signature

Solve the following application problem. Report answers to the thousandths. Include units in your answers. (4 points)

7. An archer shoots an arrow from a height of 6 feet at an angle of 25° with an initial speed of 150 ft/s.

a) Write a set of parametric equations for the motion of the arrow. (Recall:
$x = (v\cos\theta)t,\ \ y = (v\sin\theta)t - 16t^2 + h$)

b) Determine the maximum height reached by the arrow.

c) Determine how long the arrow is in the air.

d) Determine how far the arrow traveled in the air.

Write the first five terms of the sequence. (3 points)

8. $\{a_n\} = \left\{(-1)^n \dfrac{n+2}{2^n}\right\}$

A sequence is defined recursively. Write the first five terms. (3 points)

9. $a_1 = 1; \quad a_2 = 3; \quad a_n = a_{n-1} + 2a_{n-2}$

Write out the sum. (3 points)

10. $\displaystyle\sum_{k=1}^{n} \frac{k}{k+3}$

Express the sum using summation notation. (3 points)

11. $1 - \dfrac{1}{4} + \dfrac{1}{9} - \dfrac{1}{16} + \cdots - \dfrac{1}{144}$

Find each sum. Show the properties and formulas you use, then use your calculator. (#12 is 3 points and #13 is 4 points)

12. $\sum_{k=1}^{25} (4k-3)$

13. $\sum_{k=7}^{18} \left(k^2+5\right)$

Expand using the Binomial Theorem. (4 points)

14. $(x-3)^5$

Answer Key

Work on the left, the reason beside it. Boxed = final answer.

Day 2: Calculator

7. Arrow: $h = 6$ ft, $\theta = 25°$, $v = 150$ ft/s. Calculator in **degree** mode, Graph in **Parametric**.

a) $x = (150\cos 25°)t$ — *plug v and θ into the given formula; degree symbol required*

$y = (150\sin 25°)t - 16t^2 + 6$ — *+h: starts at 6 ft*

$x_1(t) = 150\cos(25°)\cdot t,\ \ y_1(t) = 150\sin(25°)\cdot t - 16t^2 + 6$ — *what you type in*

$0 \le t \le 5$, tstep 0.01; x : 0 to 600, y : 0 to 80 — *graph doesn't come back down? raise tmax*

b) $\boxed{y \approx 68.791 \text{ ft}}$ — *trace to the top ($t \approx 1.981$); not Analyze, Maximum*

c) $\boxed{t \approx 4.055 \text{ s}}$ — *trace to $y \approx 0$; trace step 0.01, then 0.001*

d) $\boxed{x \approx 551.199 \text{ ft}}$ — *same point, read x*

Other way for c): $0 = -16t^2 + (150\sin 25°)t + 6$, *quadratic formula, keep the positive t. Units: feet.*

8. $\left\{(-1)^n \frac{n+2}{2^n}\right\}$

$-\frac{3}{2},\ \frac{4}{4},\ -\frac{5}{8},\ \frac{6}{16},\ -\frac{7}{32}$

$\boxed{-\frac{3}{2},\ 1,\ -\frac{5}{8},\ \frac{3}{8},\ -\frac{7}{32}}$

reduce $\frac{4}{4} = 1$, $\frac{6}{16} = \frac{3}{8}$; *odd n negative*

9. $a_1 = 1;\ a_2 = 3;\ a_n = a_{n-1} + 2a_{n-2}$

$a_3 = a_2 + 2a_1 = 3 + 2(1) = 5$ — *previous plus twice the one before*

$a_4 = 5 + 2(3) = 11,\ a_5 = 11 + 2(5) = 21$

$\boxed{1,\ 3,\ 5,\ 11,\ 21}$

10. $\displaystyle\sum_{k=1}^{n} \frac{k}{k+3} = \boxed{\frac{1}{4} + \frac{2}{5} + \frac{3}{6} + \cdots + \frac{n}{n+3}}$

first terms from $k = 1, 2, 3$, *ellipsis, last term* $k = n$

11. $\boxed{\displaystyle\sum_{k=1}^{12} (-1)^{k+1} \frac{1}{k^2}}$

bottoms are squares $1, 4, 9, \ldots, 144 = 12^2$; *first term positive:* $(-1)^{k+1}$; $k = 12$ *gives* $-\frac{1}{144}$

12.

$4\sum_{k=1}^{25} k - 3\sum_{k=1}^{25} 1$ — *(3), (1)*

$4\left[\frac{25(26)}{2}\right] - 3(25)$ — *(6), (5); then calculator*

$1300 - 75 = \boxed{1225}$

13.

$\sum_{k=7}^{18} k^2 + \sum_{k=7}^{18} 5$ — *(2)*

$\left[\sum_{1}^{18} k^2 - \sum_{1}^{6} k^2\right] + \left[\sum_{1}^{18} 5 - \sum_{1}^{6} 5\right]$ — *(4): lower index 7, so $j = 6$*

$\left[\frac{18(19)(37)}{6} - \frac{6(7)(13)}{6}\right] + [5(18) - 5(6)]$ — *(7), (1), (5)*

$(2109 - 91) + (90 - 30) = \boxed{2078}$

14. $(x-3)^5$: row 1 5 10 10 5 1, $a = -3$

$x^5 + 5x^4(-3) + 10x^3(-3)^2 + 10x^2(-3)^3 + 5x(-3)^4 + (-3)^5$ — *blanks first; the minus goes with the 3*

$x^5 - 15x^4 + 10(9)x^3 + 10(-27)x^2 + 5(81)x - 243$ — *odd powers of -3 negative; calculator fine here*

$\boxed{x^5 - 15x^4 + 90x^3 - 270x^2 + 405x - 243}$ — *signs alternate (second term negative)*